11.1.1 Sum of Digits of a Number

A] Algorithm

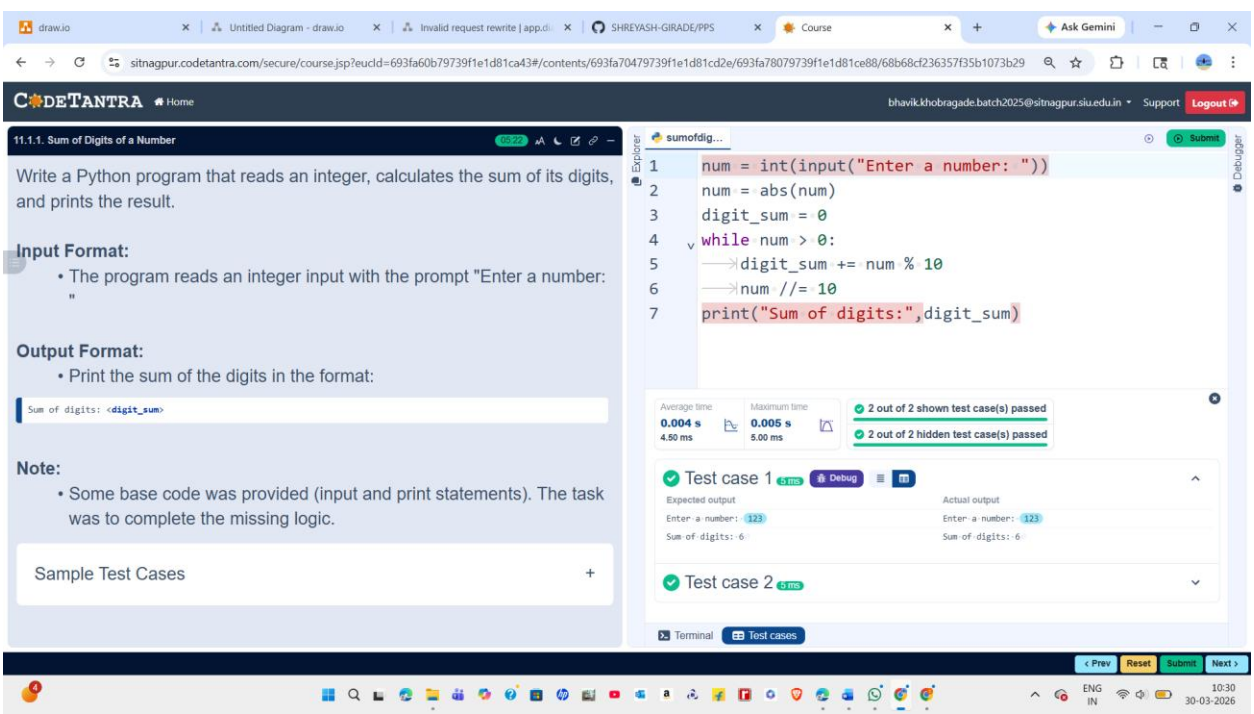
Step 1: Start
Step 2: Read an integer from the user and store it in num
Step 3: Initialize sum = 0
Step 4: While num > 0, do the following:
    Step 4a: Find the last digit of num using digit = num % 10
    Step 4b: Add digit to sum
    Step 4c: Remove the last digit from num using num = num // 10
Step 5: After the loop ends, print sum as the sum of digits
Step 6: End

B] code

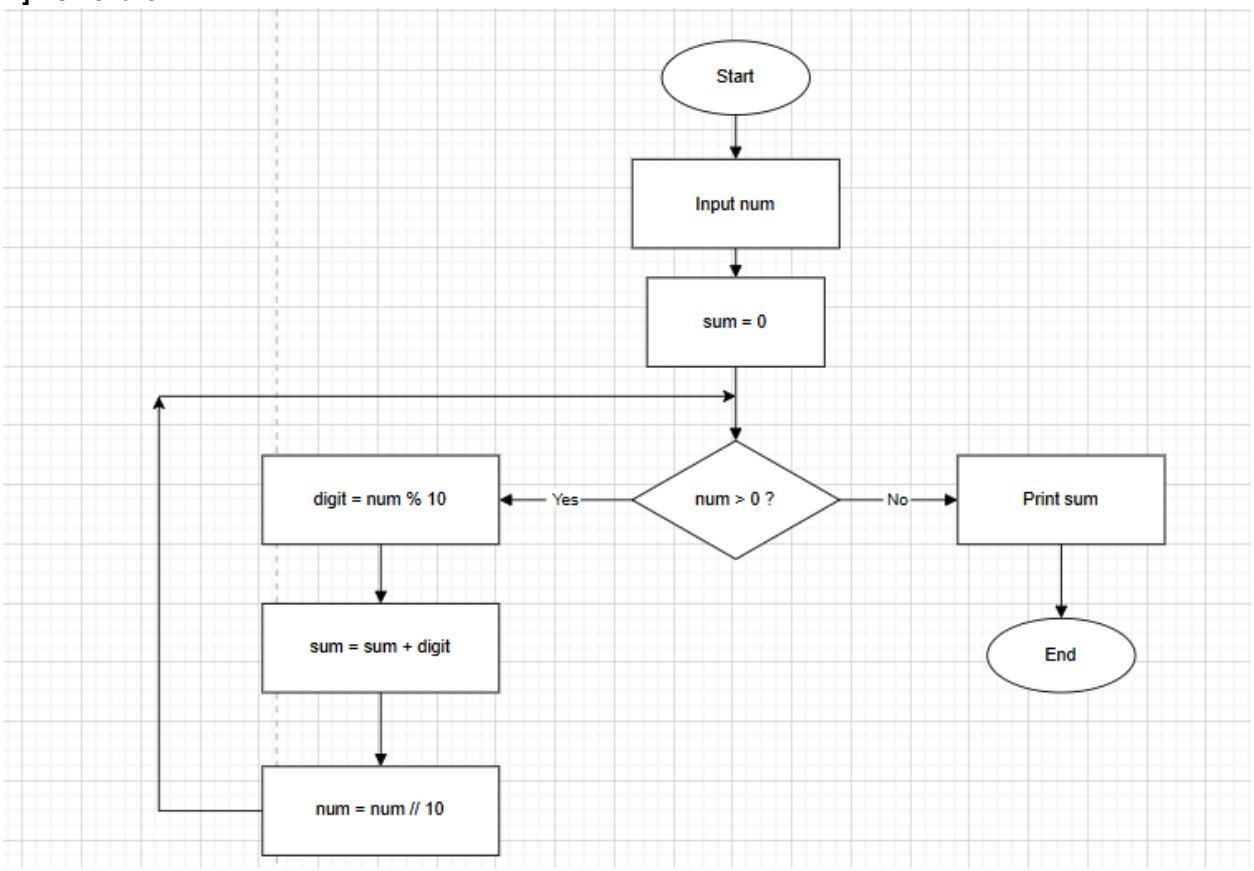
```
num = int(input("Enter a number: "))
sum=0
while num>0:
        digit = num % 10
        sum += digit
        num = num //10

print("Sum of digits:", sum)
```

C] output



D] flowchart


Start
Input num
sum = 0
num > 0 ?
Yes
No
digit = num % 10
sum = sum + digit
num = num // 10
Print sum
End